

REVIEWS

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Study on the mineral processing technology of copper-nickel sulfide ore

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Abstract

This paper briefly introduces the research status of mineral processing and mineral processing agents and summarizes the mineral processing technical difficulties of copper-nickel sulfide ore into five points: the difficulties in dissociation of copper-nickel mineral monomer, magnesium-containing silicate minerals have a great influence on flotation, the separation process of copper and nickel easily leads to the mixing of nickel and copper, effects of certain ions on flotation, and the recovery of associated precious metal elements. The key technical problems for beneficiation of copper-nickel sulfide ore are discussed and pointed out that the embedding particle size, embedding relationship, and gangue mineral type determine the formulation of the mineral processing process. Based on process mineralogy research, developing a scientific and reasonable comprehensive mineral processing process is the key to doing a good job of copper-nickel processing technology indexes, comprehensive recovery of various valuable metals, and improved economic benefits.

Keywords: Copper, Nickel, Grade, Recovery, Flotation, Mineral processing

Introduction

Nickel is an important non-ferrous metal resource, which is mainly refined from nickel ore. Nickel ore is divided into two categories: sulfide ore and oxidized ore, among which nickel sulfide ore generally contains copper, also known as copper-nickel sulfide ore. At present, the development and utilization of nickel ore are mainly nickel sulfide ore. The main countries rich in nickel ore resources are Canada, Russia, Australia, Brazil, Indonesia, China, and South Africa (Table 1). Nickel sulfide ore generally obtains nickel concentrate or copper-nickel mixed concentrate through dressing, and then undergoes pyrometallurgical processing and hydrometallurgical processing to obtain nickel metal, which is then processed into alloys or other materials. The nickel minerals in copper-nickel sulfide ore are mainly nickel pyrite and violarite, copper minerals are mainly chalcopyrite and cubanite, other sulfide minerals also include pyrite and pyrrhotine, and gangue minerals are mainly magnesium-containing silicate minerals, such as pyroxene, olivine, talc, and serpentine [1].

The mineral composition of copper-nickel sulfide is complex and diverse, among which the sulfide minerals are easy to be oxidized, the gangue mineral argillization

Table 1 World mine production and reserves (USGS)

Country	Mine production (t)		Reserves (t)
	2022	2023	
USA	17,500	17,000	340,000
Australia	155,000	160,000	24,000,000
Brazil	88,500	89,000	16,000,000
Canada	143,000	180,000	2,200,000
China	114,000	110,000	4,200,000
Indonesia	1580,000	1,800,000	55,000,000
New Caledonia	200,000	230,000	7,100,000
Philippines	345,000	400,000	4,800,000
Russia	222,000	200,000	8,300,000
Other countries	404,000	380,000	> 9,100,000
World total (rounded)	3,270,000	3,600,000	> 130,000,000

occurs easily, and the beneficiation is difficult. Therefore, it is of great significance to analyze and understand the research status and progress of copper-nickel sulfide ore mineral processing technology and mineral processing agents, to provide research ideas and countermeasures for the technical progress of copper-nickel sulfide ore beneficiation, and for the development and utilization of nickel ore resources.

World nickel ore resources, main nickel-containing minerals, and large copper-nickel concentrators

World nickel mineral resources

USGS data up to 2023, World Mine Production and Reserves: Reserves for Australia, China, Indonesia, Russia, the USA, and “other countries” were revised based on company and government reports.

World resources

Globally, nickel resources have been estimated to contain more than 350 million tons of nickel, with 54% in laterites and 35% in magmatic sulfide deposits. Hydrothermal systems such as iron-nickel alloy, sedimentary-hosted polymetallic, and volcanogenic massive sulfide deposits, as well as seafloor manganese crusts and nodules, contain 10%, and miscellaneous resources such as tailings, 1%.

Substitutes

Low-nickel, duplex, or ultrahigh-chromium stainless steels have been substituted for austenitic grades in construction. Nickel-free specialty steels are sometimes used in place of stainless steel in the power-generating and petrochemical industries. Titanium alloys can substitute for nickel metal or nickel-base alloys in corrosive chemical environments.

Main copper-nickel sulfide minerals

At present, there are about 50 known nickel-containing minerals in the world. The main nickel-containing minerals are shown in Table 2.

Table 2 The main nickel-containing minerals

Mineral name	Chemical formula	Color	Ni-content (%)
Nicopyrite	(Fe,Ni) ₉ S ₈	Bronze yellow	34.22
Millerite	NiS	Light copper yellow	64.67
Vaesite	NiS	gray	47.79
Godlevskite	Ni ₇ S ₆	Dark yellow	68.26
Hengleinite	(Co,Ni,Fe) ₉ S ₆	white	9.1
Tripartite sulfur nickel ore	Ni ₃ S ₈	Light copper yellow	40.69
Sulfur nickel ore	NiNi ₂ S ₄	Light gray to steel gray	57.86
Violarite	FeNi ₂ S ₄	Brown or purple	38.94
Quaternary pyrite (mackinawite)	(Fe,Ni,Co)S	bronze	18~22
Gersdorffite	NiAsS	Tin-white to steel-gray	35.41
Niccolite	NiAs	Light copper yellow	43.92
Water nickel ore	Ni(OH) ₂	Light green blue	63.32
Bunsenite	NiO	Dark green	78.58

Explanation

Nickel can also be present in pyrrhotite as an isomorphic impurity. The chemical formula of biotite is $K\{(Mg,Fe)_3(AlSi_3O_{10})(OH,F)_2\}$. Among them, at the positions of cations such as Mg^{2+} and Fe^{2+} , nickel (Ni^{2+}) with an ionic radius similar to that of nickel ions can partially replace them and enter the crystal lattice of biotite, forming an isomorphic substitution.

Large copper and nickel sulfide concentrator in the world

There are hundreds of copper-nickel sulfide concentrators in the world. Among them, large concentrators with relatively distinctive processes are the Norilsk concentrator in Russia that applies the combined process of gravity separation-mixed flotation-separation flotation, the Mt Keith concentrator of Western Nickel Company in Australia that adopts narrow-size beneficiation, Jinchuan concentrator in China that adopts three-stage one-closed-circuit stage grinding and stage separation, and the Klara-bel concentrator that adopts a three-stage open-circuit crushing—"magnetic separation + flotation"—silt-sand separation process (see Table 3) [2].

Production and research status of copper-nickel sulfide ore**Research status of the mineral processing process****Mixing flotation process**

In the mixed flotation process, copper and nickel sulfide minerals are usually floated as mixed concentrate under the condition of inhibiting the floating gangue minerals and then copper and nickel separation or smelting into nickel matte and then flotation separation [3].

Yibing et al. [4] through the process mineralogy study on a copper-nickel sulfide ore containing copper 0.74% and nickel 2.04% found that the associated component was simple, the content of magnesium oxide is low, and the content of pyrite is high. Under the premise of not adding conventional dispersant and mud inhibitor, three agents of sodium carbonate, butyl xanthate, and pine oil were used through the selection process

Table 3 The world's largest copper-nickel concentrator

Concentrator's name	Capacity (t/d)	Mineral processing products	Mineral processing indexes					
			Raw ore grade (%)		Concentrate grade (%)		Recovery (%)	
			Ni	Cu	Ni	Cu	Ni	Cu
Clarabelle Mine, Canada	35,000	Copper-nickel mixed concentrate	1.5	1.2	10.2	10.6	71	86
		Sulfur concentrate			1.3	0.5	22	9
Montex, Australia	35,000	Copper-nickel mixed concentrate	0.5~0.6		16.5		60	
No.1 concentrate in Norilsk, Russia	25,000	Nickel concentrate	0.4~0.7	0.5~0.9	5.2~5.8		80~85	
		Copper concentrate				24~28		73~78
Pechenga No.1 concentrator, Russia	15,000	Copper-nickel mixed concentrate	0.65	0.3	6	3	75	74
Jinchuan concentrator No.1 beneficiation workshop, China	14,000	Copper-nickel mixed concentrate	1.3	0.63	9.0	3.6	86.5	71
Kambalda, Australia	4500	Copper-nickel mixed concentrate	2.69	0.2	12.5	0.98	89	93
Kotalahti, Finland	1500	Nickel concentrate	0.78	0.31	6.83	0.81	92.5	27.9
		Copper concentrate			0.87	28.6	0.7	62.1

of one coarse, three fine, and four sweeps. The mixed concentrate with a copper grade of 1.94%, copper recovery of 97.47%, nickel grade of 5.16%, and nickel recovery of 92.13% was obtained. For a copper-nickel sulfide ore with high mud content, CMC + SHMP was selected as the combined inhibitor and mixed xanthate as the collector [5]. After the flotation process of one coarse, two sweeps, and one fine, the mixed concentrate with a grade of copper of 4.27%, copper recovery of 85.03%, nickel grade of 10.89%, and nickel recovery of 81.61% was obtained.

Acid method flotation process

Mineral pulp pH plays an important role in the flotation of copper-nickel sulfide ores. It is found that the recovery rate of copper and nickel minerals on flotation under acid conditions is significantly higher than that of neutral and alkaline conditions. The main reasons for the higher recovery rate of copper and nickel under acid conditions are that acid can prevent the surface oxidation of sulfide minerals and increase its floating [6]; acid can also reduce the influence of gangue ore mud on useful mineral flotation [7].

Priority flotation process

Direct priority flotation process or partial priority flotation method. This method can be used when the copper content in the ore is higher than the nickel content. Its advantage is that the copper concentrate with lower nickel content can be obtained directly. The disadvantage is that the nickel mineral is inhibited when the copper float is not easily activated to affect the nickel recovery rate. Therefore, few mines use the priority flotation process alone in industry [8]. Xuejun et al. [9] used the “partial priority floating copper” → “copper-nickel mixed floating” → “copper-nickel separation” process to treat

a high copper and low nickel sulfide ore, and the technical indicators obtained are as follows: the composite copper concentrate contains 32.26% of Cu, with a Cu/Ni ratio of 47.44%, and the copper recovery rate reaches 91.66%; the nickel concentrate contains 4.60% of Ni, with a Ni/Cu ratio of 11.22%, and the nickel metal recovery rate is 80.63%.

Flash-speed flotation process

Flash flotation is a method of flotation recovery for the separation of coarse minerals in the process of grinding classification. It is completed by using a special flotation machine, which is in line with the principle of “can receive early harvest” in the beneficiation process. The application of the flash flotation process can effectively improve the recovery rate of mineral processing, reduce the cycle load of the grinding circuit, and reduce the reagents consumption, which is widely used in copper-nickel sulfide ore, lead-zinc sulfide ore and gold and silver ore. Yuanke [10] introduced the flash flotation industrial test conducted by the Jinchuan nickel dressing plant. Compared with no flash flotation machine, the recovery rate of nickel and copper was increased by 1.32% and 0.75%, respectively; the particle size content of $-74\ \mu\text{m}$ was reduced by 32%; the dosage of flotation agents and the dehydration cost of concentrate concentration were reduced; and the economic benefits were improved.

Stage grinding—stage separation process

The mineral mosaic distribution relationship in copper-nickel sulfide ore is complex and usually requires fine grinding to realize monomer cleavage. However, fine grinding leads to the production of a large amount of ore mud, which affects the flotation of sulfide minerals. Using stage grinding and stage separation processes can reduce ore overcrushing [11]. In the process of Central South University's flotation study of rich ore in the Jinchuan Second Mining area [12], the good selection index was obtained by using the process of two grinding and two dressing and regrinding of medium coarse concentrate.

Li et al. [13] believed that after comparing and analyzing the indicators before and after the application of the separate regrinding and re-election process of middlings, the two-stage grinding and two-stage separation process, and the “two-stage grinding and two-stage separation + regrinding of middlings” process, the two-stage grinding and two-stage separation process is superior to the separate regrinding and re-election process of middlings in terms of both the total concentrate nickel grade and the recovery rate index. The nickel grade of the total concentrate of the former reaches 7.96%, which is 0.26% higher than that of the latter. The nickel recovery rate reaches 83.65%, which is 0.25% higher than that of the latter. After comparing the “two-stage grinding and two-stage separation” process and the “two-stage grinding and two-stage separation + regrinding of middlings” process, it is found that the latter can greatly improve the low concentrate nickel grade, from 3.95% before implementation to 4.45% after implementation. At the same time, the nickel grade of the total concentrate is increased from 8.061 to 8.28%.

Magnetic flotation process

Magnetic flotation is a process of completing magnetic separation and flotation in one device at the same time, which prevents magnetic minerals from entering into the

foam product during non-magnetic mineral flotation. Magnetite ore and nickel ore can be treated by this method. Yalxin et al. [14] proposed to use the “magnetic flotation method” to effectively treat the ore containing pyrrhotine. This method reduces the content of pyrrhotine in flotation concentrate by applying the magnetic field to leave the pyrrhotine in the flotation tank. The mineral processing index of this method is much better than that of the conventional flotation method. The biggest advantage of magnetic flotation is that it combines two sorting processes into one process, which can reduce the number of intermediate material processing and simplify the process flow. When applying magnetic flotation, only one process needs to be optimized, so the optimization work is simple and effective.

Distinguishing speed flotation process

According to the difference in flotation rate, the mineral particles in the pulp can be divided into three categories: the first category is the mineral particles with floating up between 17 and 34 min, whose flotation rate is very slow, called “slow particles,” Ni with low grade and magnesium-containing silicate gangue exists in large quantities; the second category is the mineral particles with floating rate high, called “fast particles,” high Ni grade, and low MgO content; the third category is the mineral particles with floating up between 7 and 17 min, called mid-speed particles. In order to realize the early harvest and fast harvest of useful sulfide minerals, reduce the pollution of mineral sludge as far as possible, and obtain high-quality concentrate with low magnesium content, the distinguishing speed flotation process should be adopted [15].

Research status of mineral processing agents

A flotation agent is very important for the flotation process, which can greatly increase the hydrophilic difference between the target mineral and gangue mineral surface, thus bringing favorable conditions to flotation operation. The effect of flotation agents has been a key factor restricting flotation operation for a long time. The commonly used nickel sulfide flotation agents have disadvantages such as poor harvesting effect and poor selectivity [16], and more targeted new nickel sulfide flotation agents have been continuously developed and applied.

Collector

Commonly used copper-nickel sulfide ore collectors include two categories, one is anionic collector, such as xanthate, black drug, and sulfur-nitrogen; this kind of agent is partially soluble in water. Ionization produces anions containing sulfur atoms that have capture capacity for copper-nickel sulfide ore; one is a lipid agent with polarity, cannot ionize in water, and is a derivative of xanthate, black drug, and sulfur-nitrogen, such as double xanthate, xanthate ester, double black drug, and black drug ester. Usually, the latter category has better capture capacity than the former, but the former is more selective than the latter [16].

Amide alcohol xanthate is an amphoteric collector that is several times better for many metal sulfide mines than conventional collectors. Under acid pH, it will produce electrostatic repulsion with nicopyrite, and its harvesting ability is relatively weak; however, under neutral and weak alkaline pH conditions. Due to electrostatic and

even chelating adsorption on the surface of nicopyrite, the harvesting ability is greatly enhanced, which is better than butyl xanthate. In the flotation of copper-nickel sulfide ore in the second mining area of Jinchuan by using amide alcohol xanthate as the collector, the grade of concentrate product nickel is 7.60%~8.27%, and the nickel recovery rate is 88.52%~89.13%. Compared with the use of butyl xanthate, the grade of concentrate nickel can be increased by 1%~2% [17].

With the leanness, fineness, and complexity of copper-nickel sulfide ore, some collectors with strong harvesting ability and good selectivity have been developed continuously, and some still have both foaming properties. BF-4 developed by the pharmaceutical factory of Baiyin Company has the characteristics of fast flotation speed, brittle foam, and small dosage. Compared with the use of butyl xanthate, the recovery rate of concentrate nickel and copper increased by 0.59% and 1.58% respectively, and the content of magnesium oxide decreased by 0.86% [18]. “Y89 + PN405” jointly developed by Guangzhou Nonferrous Metals Research Institute and Zhuzhou Pharmaceutical Factory is an effective combination of harvesting agent and foaming agent, and Y89 series xanthate is a new high-efficiency collecting agent; PN405 is a new high-efficiency collecting-bubble agent, copper selection, and harvesting ability are strong and has a certain foaming ability. In the flotation test of the second mining area of Jinchuan Company, the combined prescription can improve the recovery rate and reduce the magnesium oxide content by ensuring that the grade of nickel and copper is comparable to the on-site pharmaceutical system [19]. BS-4 developed by Feng Qiming and Zhang Guofan [20, 21], J-622 developed by Northwest Institute of Mining and Metallurgy [22, 23], and C-125 developed by Dai Shujuan [24] have also achieved very good harvesting effect.

Bubble agent

The traditional foaming agent is pine camphor oil, which is also the most common and widely used foaming agent in flotation. However, copper and nickel ores are generally produced in ultra-basic rocks or basic rocks, containing a large amount of ophiolite, amphibole, talc, and other magnesium silicate, which results in large ore mud in the flotation process, high foam viscosity, and obvious entrainment of gangue, which is not conducive to mineral sorting. Therefore, it is very important to study highly targeted and selective foaming agents [25].

BK-206 is a high-efficiency foaming agent made from petrochemical products, which mainly contains advanced fatty alcohols and ether ester compounds. It has a strong foaming capacity, the amount can be greatly reduced, and the formation of foam is very stable. The nickel concentrate recovery and MgO content were slightly improved [26]. The disulfide formic acid-2-cyanoethyl ester developed by Xue Dingshuai has a strong ability to foam and collect, and its selectivity is better than the sulfur nitrogen collector. In the closed-circuit flotation test of Jinchuan copper-nickel sulfide ore, the use of this agent can improve the grade and recovery rate of concentrate nickel and copper compared with the original agent, but the effect decreases in winter [22].

Gangue inhibitor

In the copper-nickel sulfide ore flotation test study, inhibitors are mainly divided into inorganic inhibitors and polymer organic inhibitors. Polymer inhibitors are mainly for

glycan natural organic matter and its by-product [1]. Domestic use of inhibitors mainly such as Calgon, sodium silicate, salufer, sodium pyrophosphate, CMC, and dextrin [27]. Changhuai et al. studied the inhibitory effect of dextrin, sodium silicate, Calgon, and CMC on ophiolite [28]. In recent years, many scholars have also developed and synthesized inhibitors expressed by code names, most of which are compound combinations of compounds with known structures, such as EP and JCD. There are few inhibitors of new structures, and the exploration and use of combined agents need long-term experimental research.

The worldwide researches of magnesium-containing silicate mineral inhibitors focus on organic matter inhibitors, with in-depth theoretical research and many varieties, but carboxymethyl cellulose (CMC) is widely used in industry. Guar bean gum is widely used in the South African mining industry as an inhibitor of silicate gangue minerals such as talc in sulfide mine flotation, which has very good results [29]. Makarinsky [30] used Gubean gum as an inhibitor in the flotation of copper-nickel sulfide ore. Under the condition that raw nickel and copper grades are 0.50% and 0.76%, the nickel and copper grades of copper-nickel mixed concentrate are 4.42% and 8.50%, respectively, and the recovery rate of nickel and copper is 67.0% and 89.3%, respectively [31, 32].

Danning[33] conducted research on flotation inhibitors for copper-nickel sulfide ores and synthesized a binary copolymer (N-MAAA) with chelating ability and several other macromolecular polymers containing carboxylic acid groups. The flotation experiment results of Jinchuan copper-nickel sulfide ore show that adding sodium polyaspartate (PASP) and epoxy succinic acid polymer (PESA) can effectively improve the total recovery rate of copper and nickel compared with on-site conditions. The binary copolymer N-MAAA can not only improve nickel-copper flotation indicators but also reduce the magnesium oxide content in the second-stage concentrate from 9.02 to 7.59%. Therefore, N-MAAA is a potential selective inhibitor for the flotation of copper-nickel sulfide ores.

In order to eliminate the oxide film on the surface of copper-nickel sulfide minerals and suppress the influence of magnesium-containing gangue slime such as serpentine on flotation, based on the acidic flotation process of copper-nickel sulfide ores, Hu Cong et al. [34] studied the optimization effect of carboxymethyl cellulose (CMC) on the flotation process of copper-nickel sulfide ores and discussed the mechanism of action of CMC. The results show that 1×10^{-3} mol/L hydrochloric acid solution and 20-min ultrasonic pretreatment can significantly enhance the floatability of copper-nickel sulfide ores. The acidic flotation system can achieve better recovery rates in a wider pH range. Adding 20 mg/L CMC in the acidic system with $\text{pH} = 3$ can increase the nickel grade of copper-nickel sulfide ore flotation concentrate to 19.42% and the recovery rate to 60.05%. Compared with the flotation index without adding CMC under the same conditions, it increased by 1.78% and 22.79%, respectively.

Adjustment agent

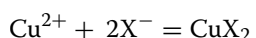
The flotation of copper-nickel sulfide ore can be performed at acidic, neutral, and weakly alkaline pH values. Sodium carbonate is commonly used to adjust the alkaline pH pulp. After sodium carbonate dissolves, the pulp can obtain a certain pH buffer effect and

reduce the Ca^{2+} , Mg^{2+} , and other inevitable ions in the pulp. In addition, sodium carbonate also has a certain role in dispersing fine mineral sand [3].

Acid pH pulp is commonly used to regulate ammonium sulfate, which has a certain cleaning effect on the oxide film on the surface of copper-nickel mineral particles so that its active surface is exposed to facilitate the interaction of surface metal ions and collectors. Qiang [35] took Xinjiang Kalatunk copper-nickel sulfide mine as the research object and obtained high-quality mixed concentrate with a copper grade of 6.75%, a nickel grade of 3.44%, a copper recovery rate of 93.15%, and nickel recovery of 82.33% under the environment of weak acid pH.

Activator

At present, copper-nickel mines mainly use copper sulfate to activate nicopyrite in China, and Cu^{2+} can form copper sulfur compound film on its surface to increase its surface hydrophobicity and then improve the flotation index. When using copper sulfate, attention should be paid to its amount. When the amount is too large, the excess Cu^{2+} in the pulp will react with the xanthate ion and consume the agent but will reduce the flotation recovery rate [36]. The reaction type is.



In addition, because the positively charged Cu^{2+} and copper-containing hydroxyl components (such as CuOH^{+}) cover the surface of gangue minerals, it will activate some gangue minerals to a certain extent, which is unfavorable for the selection effect [37, 38]. Junwei [39] and Xuanwu [40] found that EDTA disodium and citrate acid could effectively weaken the activation of some magnesium-containing silicate minerals by complexing mineral pulp Ni^{2+} and Cu^{2+} and had a significant quality improvement and magnesium reduction effect.

Yang et al. [41] studied the removal effect of walnut shell-based activated carbon on these residues in view of the situation that metal ions and residual organic agents in copper-nickel bulk flotation pulp will affect the separation effect of copper and nickel. The research shows that walnut shell-based activated carbon $\text{H}_2\text{SO}_4/\text{AC}$ has strong adsorption capacity for Cu^{2+} , Mg^{2+} , and sodium ethyl xanthate, and the adsorption process satisfies the quasi-second-order kinetic equation. In flotation tests, compared with conventional activated carbon, $\text{H}_2\text{SO}_4/\text{AC}$ can significantly improve the separation effect of copper and nickel. The nickel grade in copper concentrate is reduced from 4.25 to 0.22%, and the copper grade in nickel concentrate is reduced from 0.66 to 0.21%. In addition, the recoveries of copper and nickel also increased by 8.21% and 4.56%, respectively.

Technical difficulties in beneficiation of copper-nickel sulfide ore

Difficulty in dissociation of copper-nickel mineral monomer

Due to the internal composition of copper-nickel sulfide ore being extremely complex, there are as many as four or five metal minerals, such as chalcopyrite and nicopyrite. The sulfide has a natural floating property [42] and excellent performance, but uneven mosaic density between the internal mineral particles, monomer dissociation between inadequate and incomplete easy to cause the final excessive grinding powder phenomenon. Nickel minerals are mainly distributed in nicopyrite, but nickel minerals are

closely embedded with metal sulfide ores such as pyrrhotine, resulting in a part of the nicopyrite being present in the crystal lattice of pyrrhotine in the form of isomorphism. However, even in this case, fine grinding is difficult to fully dissociate nickel, and the nicopyrite enters the tailings very easily with other impurity minerals, thus reducing the recovery rate of nickel. In addition, some chalcopyrite and other ores are highly likely to form closer mosaic density after mixing with them and may lead to more refined density between ores, which will lead to more difficulty of copper and nickel mineral monomer dissociation [43, 44].

Magnesium-containing silicate minerals have a great influence on flotation

Serpentine is the main magnesium-containing gangue mineral in the flotation separation system of copper-nickel sulfide ore. The chemical composition of serpentine $[\text{Mg}_6(\text{Si}_4\text{O}_{10})(\text{OH})_8]$ is mainly MgO (43.6%), SiO_2 (43.3%), and H_2O (13.1%). It is formed by the serpentine of olivine and pyroxene in the rocks at medium–low temperature metasomatism or medium–low region action. Its hardness is 2 ~ 3.5, and its relative density is 2.5 ~ 2.62 [45].

Jianqing et al. [46] believe that the serpentine argillization occurs very easily. The main characteristics of fine-grained serpentine are small mass, large specific surface area, and high surface energy. The small mass causes the small momentum of the ore in the pulp and the small collision probability with the bubble, so it is difficult to overcome the energy barrier between the ore particles and the bubble and adhere to the surface of the bubble. However, once the serpentine ore particles adhere to the bubble surface, it is difficult to fall off. This fine gangue mineral is stuck by the water medium, which is easy to rise into the foam layer with the water flow to form inclusion. Large specific surface area and high surface energy cause serious non-selective agglomeration between serpentine gangue mineral particles, and the “self-condensation” phenomenon of ore mud often occurs. It is easy to form a cover on the useful mineral surface; the adsorption increases, and the selectivity of adsorption decreases; the solubility of mineral particles in a water medium increases, and the inevitable ions in the pulp increase; the factors that are not conducive to flotation, such as the large increase of ore pulp viscosity.

The separation process of copper and nickel easily leads to the mixing of nickel and copper

The copper-nickel separation method is to separate the copper-nickel crystal ore obtained by flotation technology again and obtain two high-purity concentrates: nickel concentrate and copper concentrate [47]. In the whole process, the flotation technology can be used to directly extract the copper element or the copper-nickel mixed ore can be smelted into nickel matte material, and then, the nickel matte is highly polished and high flotation to separate and emerge. Problems occur throughout the separation process due to technical problems. For example, due to the symbiotic coexistence of copper and nickel elements in copper-nickel ore, the chimerism density between copper and nickel is small, and the two material elements not completely separated are very easy to adsorb into each other, resulting in the reduction of the separation and recovery efficiency of nickel elements.

Effects of certain ions on flotation

The essence of conventional copper and nickel separation is the separation of copper and sulfur, that is by using the lime method. The main defect of this method is that it must be carried out in a high alkali ($\text{pH} \geq 12$) medium. The problem caused by the flotation of a high alkali medium is that the flotation operation is difficult [48]. After grinding under normal pH, some ions can change the flotation behavior of minerals: for example, when the battery effect of iron in the mill works, calcium ions can activate the flotation of copper sulfide and nickel ore. Because the slurry is alkaline, there are copper, nickel, and iron ions present in the liquid phase of the slurry. In the process of grinding and flotation, a redox reaction occurs, and the metal complex is formed with thiosulfate ions, thus affecting the flotation index. Liu [49] studied and found that the presence of Fe^{3+} had a significant inhibitory effect on Cu_2S and Ni_3S_2 ore and had a good correspondence with the slurry electric potential. The presence of Cu^{2+} activates Ni_3S_2 ore flotation and interferes with copper-nickel separation. The presence of Ni^{2+} has little effect on Ni_3S_2 mine flotation, but when its dosage is large, Cu_2S mine flotation may be affected by the consumption of xanthate.

Recovery of the associated precious metals

In copper-nickel sulfide ore, there is often a certain amount of gold, silver, and platinum group elements such as platinum, palladium, osmium, and iridium precious metals. The presence state of these elements is highly complex, but they are mostly enriched in nickel concentrate in the process of flotation. Therefore, the separation of some copper concentrates that basically contains none of these elements from the mixed concentrate is conducive to the enrichment and recovery of these precious metal elements. In particular, the process of recovering precious metals from smelting intermediates-nickel matte is even more so. In the slow cooling of nickel matte, in the differentiation products of the different components occurred, the yield of the alloy phase (nickel-copper-ferrum rich in precious metal) is related to the content of copper in nickel matte. If the content of copper is high, the alloy phase yield is also high, and the enrichment ratio of precious metal is low [48].

Discussion on several key technical problems of copper-nickel sulfide ore mineral processing

Through process mineralogy studies, the common characteristics of copper-nickel sulfide ore are gangue minerals are mainly serpentine and talc, high MgO content, brittle, and argillization arise easily, which are easily activated by grinding to produce Ni^{2+} and Cu^{2+} ; natural floatability is good, and flotation inhibition is difficult; the particle size of the sulfide mineral aggregate is not uniform; the symbiotic relationship between minerals is close and extremely complex; and fine useful minerals are not easy to dissociation [50]. Copper and nickel minerals are not only easy to crush, seriously argillization and oxidation but also difficult to separate from pyrrhotine and pyrite due to their similar natural flotation, so it is difficult to obtain high-grade copper-nickel sulfide concentrate. It is precisely because of these characteristics of

copper-nickel sulfide ore that lead to the difficulty of ore flotation technology [51]. Exploring and solving these problems is exactly what we need to focus on.

Strengthening the inhibition research of MgO-containing minerals

For the flotation of copper-nickel sulfide ore rich in magnesium-silicate gangue minerals, how to reduce the content of concentrate MgO has always been the focus of research at home and abroad [52]. MgO gangue minerals have the characteristics of a high melting point, so the high MgO content in the floating concentrate of copper-nickel sulfide ore not only affects the concentrate grade but also causes excessive slag phase viscosity in the subsequent smelting operations, leading to the furnace nodules, slag phase separation difficulties, and reduced smelting recovery rate, thus increasing the production cost. Therefore, the content of MgO in the concentrate must be strictly controlled [53]. Different paths of MgO-containing gangue minerals into flotation concentrate and corresponding solutions:

- (1) Due to the uneven particle size of embedding and the difficulty of monomer dissociation, MgO gangue mineral and sulfide mineral are in the form of the complex into concentrate [54]. By optimizing the grinding and flotation process conditions, we can improve the dissociation degree of the target mineral monomer and reduce the production of fine mineral sand, thus improving the flotation environment [55].
- (2) In the process of flotation, the argillization fine gangue enters the concentrate by covering the surface of the sulfide mineral particles and the inclusion of the target mineral flocs through the way of electrostatic and magnetic adsorption. To change the surface electrokinetic potential of gangue minerals and useful minerals by adding appropriate adjustment agents, reduce the electrostatic cap adsorption of gangue minerals on useful minerals. Mechanical inclusion [53, 55] due to hydrophobic agglomeration of useful minerals can be reduced by adding dispersant and improving flotation stirring conditions.
- (3) Magnesium-containing silicate minerals such as serpentine and talc have good natural floatability and can easily enter the concentrate foam layer during flotation; Cu^{2+} and Ni^{2+} produced in the grinding process activate the magnesium-containing minerals and thus float into the concentrate layer by acting with the collector [56]. Metal-activated ions on the gangue mineral surface were removed by adding detergent and complex agents, while selective inhibitors were used to weaken the floatability of gangue minerals and reduce the possibility of entering the concentrate [54].

Chao et al. [57] carried out beneficiation test research on a low-grade sulfide copper-nickel ore abroad, which has difficulties in beneficiation such as the close dissemination relationship among pentlandite, chalcopyrite and pyrrhotite, complex symbiotic relationship, and high content of magnesium-containing silicate minerals. The results show that for the raw ore with a nickel grade of 0.50% and copper grade of 0.20%, by adopting the process of rough grinding of raw ore-fine grinding of middling-bulk flotation of copper and nickel, through selective grinding and adding high-efficiency inhibitor CMC, overgrinding of copper and nickel minerals is avoided and selective inhibition of

magnesium-containing silicate minerals is achieved. In the closed-circuit test, a copper-nickel concentrate with a nickel grade of 9.70%, copper grade of 4.75%, and nickel and copper recovery rates of 68.99% and 79.85%, respectively, is obtained, realizing the effective recovery of copper and nickel resources.

Strengthening study on the separation of copper-nickel sulfide minerals and other sulfide minerals

Although the specific mineral composition of copper and nickel sulfide deposits around the world is different, the mineralogy properties of useful minerals are roughly similar; that is, mainly nicopyrite, chalcopyrite, and pyrrhotine, which are characterized by sulfide minerals with stable mineral crystal lattice composition and good natural floatability. Violarite, chalcopyrite, pyrite, and other minerals constitute the copper-nickel sulfide deposit formed in the process of geological shallow alteration, which is characterized by unstable mineral crystal lattice composition, mineral cleavage development, easy-to-be-oxidized, and easy-to-overcrushing [58]. The flotation practice shows that among the three kinds of minerals, chalcopyrite floatability is the best, but from weakly acidic to alkaline medium, the floatability of nicopyrite is always superior to pyrrhotine; in terms of flotation dynamics, the flotation speed of chalcopyrite, nicopyrite, and violarite is higher than that of pyrrhotine and pyrite. With the continuous development and utilization of resources, the copper-nickel sulfide ore resources with stable and single mineral components, low alteration degree, and easy flotation separation of various sulfide minerals are constantly decreasing. Now, the main development and utilization of primary sulfide minerals and the staggered coexistence of secondary minerals make the mineral composition very complex, with a variety of copper-nickel sulfide ore resources that have difficulties in flotation separation of useful minerals. Therefore, it is one of the research difficulties for copper-nickel sulfide with the pyrrhotine and pyrite to flotation separate and reduce the smelting cost, improving the grade of copper-nickel concentrate [59].

Strengthening the research on new agents for beneficiation

In addition to various new agents, the commonly used copper-nickel sulfide ore flotation agents are listed in Table 4.

At present, the application of copper-nickel sulfide ore in flotation agents mainly includes several aspects: (1) efficient and selective collectors of copper-nickel sulfide ore minerals, (2) effective inhibitors of copper-nickel sulfide ore gangue minerals, and (3)

Table 4 Flotation reagents of copper-nickel sulfide ore

Type of reagent		Reagent name or chemical formula
Regulator	pH modifier	(NH ₄) ₂ SO ₄ , Na ₂ CO ₃ , NaOH, etc
	Slime depressant	Na ₂ SiO ₃ , CMC, (NaPO ₃) ₆ , Na ₄ O ₇ P ₂ , starch, Guar gum, Dextrin, etc
	Nickel activator	CuSO ₄ , Na ₂ S, etc
	Nickel depressant	CaO, NaCN, Na ₂ SO ₄ , etc
Collector		Advanced xanthate, Aerofloat, Z-200, Y-89, J-622, etc
Frother		MIBC, Pine oil, etc

there are also some attempts to selectively treat fine particles of copper-nickel sulfide by chemical leaching, microbial leaching, or other new processes, but the research on flotation agents is not ideal. In the new process, due to its own limitations and large input cost, it cannot be applied on a large scale [59].

Strengthening the research and development of new mineral processing agents, especially for other chemical, medical, environmental, and other industries with dispersing properties and with traditional agent Calgon, CMC having similar functional group agents can be used to modify or try to combine with traditional agents, to reduce the content of MgO gangue in concentrate, and explore suitable selective. Foaming efficient collectors have become an important development trend of copper-nickel sulfide ore mineral processing agents.

Bowen et al. [60] aimed at the situation that the Jinchuan copper-nickel sulfide ore has a high content of magnesium-containing silicate gangue, and after argillization, it affects the enrichment of nickel and copper metals in concentrate through heterogeneous condensation and mechanical entrainment. Taking the second-stage flotation feed of a production workshop in Jinchuan with a nickel grade of 0.42% and a copper grade of 0.32% as the research object, a suitable new reagent system in an acidic medium was studied. The test results show that under the acidic flotation condition of $\text{pH} = 4$, using sodium isobutyl xanthate as the collector, Z-200 as the auxiliary collector and terpeneol as the foaming agent, after one roughing, two cleanings, and two scavengings, a concentrate product with a nickel recovery rate of 58.92%, a copper recovery rate of 45.60%, a nickel grade of 3.54% and a copper grade of 2.03% can be obtained. Compared with the on-site reagent system, the recovery rates of nickel and copper in concentrate are increased by 7.54% and 24.40% respectively, and the content of MgO is reduced by 0.94%. The new reagent system comprehensively improves the comprehensive utilization rate of Jinchuan copper-nickel sulfide ore resources.

Strengthen the research on the recovery of low-grade copper sulfide minerals

As high-grade ore resources are gradually depleted, accelerating the exploration of efficient processing technologies for low-grade minerals has important practical significance. Wenbo [61] used Jinchuan low-grade copper-nickel sulfide ore as raw material and added sulfur, carbon powder, and sodium carbonate. The stepwise sulfidation roasting process was adopted to convert Co, Cu, and Ni existing in the oxidation phase and silicate phase in the ore into the sulfide phase. Within the range of 0–1000 °C, the sulfide phases of Co, Cu, and Ni can exist stably. Increasing sulfur partial pressure and reaction temperature is conducive to the occurrence of MeS (Me = Co, Cu, Ni). During the stepwise sulfidation roasting process, NiO, CoO, and CuO preferentially react with carbon powder and sulfur to generate MeS (Me = Co, Cu, Ni). Sodium carbonate can react with the silicates of Co, Cu, and Ni, destroy the silicate structure, generate the corresponding oxidation phase, and then improve the sulfidation rate of valuable metals through a sulfidation reaction. (3) Process conditions such as the additional amount of sulfur, carbon powder and sodium carbonate, roasting temperature, roasting time, and mineral particle size can regularly affect the sulfidation rates of Co, Cu, and Ni. During the stepwise sulfidation roasting process, Fe, Ni, and Co form stable sulfide solid solutions at high temperatures, which is conducive to the enrichment of valuable metals.

Yang et al. [62] conducted flotation test research on a low-grade, high-mud, and high-magnesium copper-nickel sulfide ore in a copper-nickel mine in Xinjiang. The process involves using sodium carboxymethyl cellulose and water glass as a combined depressant, butyl xanthate and ammonium dibutyl dithiophosphate as a combined collector, and adopting the copper-nickel bulk flotation-copper-nickel separation process flow. In the closed-circuit test, a copper concentrate with a copper grade of 25.11%, a copper recovery rate of 79.90% and containing 2.02% magnesium oxide, as well as a nickel concentrate with a nickel grade of 6.98%, and a nickel recovery rate of 75.01% and containing 5.32% magnesium oxide can be obtained. This realizes the efficient recovery and effective separation of copper and nickel.

Explore new ways to improve the beneficiation index of copper-nickel sulfide ore

As the main extraction mineral of nickel metal, the separation difficulty of copper-nickel sulfide ore is also getting higher and higher. Therefore, it is necessary to continuously explore new ways to recover copper-nickel sulfide minerals. A large number of practical results show that increasing the yield of easily floatable particle size in grinding products can effectively improve the flotation index of copper-nickel concentrate, and the method is relatively simple to operate and the improvement effect is obvious. According to the ore mechanical properties of Huangshan copper-nickel sulfide ore in Yakes, Xinjiang, the particle size characteristics of grinding products at various stages, and the flotation concentrate index and other conditions, combined with the grades of nickel and copper in the raw ore established linear regression models between the copper recovery rates of nickel and copper concentrates and multiple factors of grinding by using statistical regression analysis methods [63]. Thus, through the variation range of the yield of easily floatable particle size in grinding products, the variation range of flotation indexes can be predicted, providing prerequisite conditions for the production index control of the concentrator.

For the low-grade copper-nickel ore of talc type in Danba, Sichuan, Wei [64] carried out an industrial test on a scale of 750 tons per day by using a new inhibitor MKG on the basis of laboratory tests and expanded continuous tests. Ideal technical indicators were achieved, the subsequent test phenomena were good, and the indicators were stable. The nickel recovery rate of mixed flotation reached 83.38%, and the copper recovery rate reached 90.16%. Compared with the original “desliming-flotation” process at the production site, the nickel recovery rate was increased by 16.96%, and the copper recovery rate was increased by 36.64%. After the separation of copper and nickel, the nickel grade of the nickel concentrate is 6.36% and the copper content is 0.60%. The nickel recovery rate is 82.02%. The copper grade of the copper concentrate is 24.60% and the copper recovery rate is 72.96%. The overall technical indicators are very excellent.

Based on optical microscope analysis, Wei et al. [65] systematically analyzed the products in the beneficiation process of a copper-nickel sulfide ore by combining scanning electron microscopy, X-ray energy spectrum, X-ray diffraction, ICP, and chemical phase analysis, including the product mineral compositions and the monomer dissociation of main minerals in the primary and secondary run-of-mine ores, and primary and secondary concentrates and tailings. The results show that about 10% of nickel and 5% of copper elements in the raw copper-nickel ore exist in an oxidized state. Trace amounts

of nickel in the sulfide phase and relatively more copper in the sulfide phase are distributed in bravoite and melanocerite and are difficult to be collected by flotation. The magnesium oxide in the concentrate mainly comes from serpentine, olivine, talc, chlorite, tremolite, and diopside. Relatively coarse particles cannot be effectively dissociated from sulfides. Controlling the monomer dissociation of magnesium oxide in relatively coarse-grained grades and the floating of monomer minerals and controlling the entrainment and collection of fine-grained magnesium oxide minerals are the keys to reducing the magnesium oxide in concentrates.

Fulan et al. [66] conducted a process mineralogy study and beneficiation test research on a certain copper-nickel sulfide ore in Xinjiang. In the raw ore, minerals are unevenly distributed, with varying particle sizes and complex dissemination relationships, and gold is associated. A process flow of bulk flotation of copper and nickel-separation of copper and nickel was adopted. The new reagent ester-2 + isoamyl xanthate was selected as the collector for bulk flotation, and lime + T-15 was used as the inhibitor for nickel minerals in the copper-nickel separation. A copper concentrate with a copper grade of 22.36%, a copper recovery rate of 65.33%, a gold content of 115.00 g/t, and a gold recovery rate of 63.54%, and a nickel concentrate with a nickel grade of 6.11% and a nickel recovery rate of 71.25% were obtained [67].

In a low-grade copper-nickel ore in Guangxi, the content of magnetite is 5%, and it also contains noble metals such as silver and cobalt with relatively high grades. In view of the properties of this ore and combined with the actual conditions, Bin et al. [68] adopted a combined process of "flotation-magnetic separation". For the bulk flotation of copper and nickel, a process flow of one roughing, two cleaning, and three scavenging was used, and conventional flotation reagents were applied. Relatively ideal indexes were obtained: the yield of copper-nickel concentrate is 1.83%; the grades are 4.81% of copper, 7.34% of nickel, 0.31% of cobalt, and 230.60 g/t of silver; the recoveries are 88.69% of copper, 80.12% of nickel, 55.76% of cobalt, and 78.75% of silver. The flotation tailings are subjected to magnetic separation to recover magnetite, and finally, the yield of iron concentrate is 4.78%, the iron grade is 60.65%, and the iron recovery is 28.70% [67].

Zhiqiang [69] conducted flotation tests on the Hami copper-nickel raw ore containing 0.30% copper and 0.42% nickel using a combined inhibitor of sodium hexametaphosphate + gum arabic. Under the condition that the grinding fineness of $-74\ \mu\text{m}$ accounted for 74%, single-factor condition tests for bulk flotation-separation of copper and nickel, open-circuit and closed-circuit tests of the whole process were carried out on the raw ore. Finally, a copper concentrate with a copper grade of 21.23%, a recovery rate of 75.54%, and an MgO content of 4.02% was obtained, as well as a nickel concentrate with a nickel grade of 6.90%, a recovery rate of 74.88%, and an MgO content of 5.93%. The copper grade in the copper concentrate is greater than 20% and less than 25%, meeting the third-level standard for copper concentrates; the nickel grade in the nickel concentrate is greater than 6.5% and less than 7.5%, meeting the fourth-level standard for nickel concentrates.

Feng et al. [70] conducted a study on a certain copper-nickel ore in Xinjiang, which contains 0.33% Cu, 0.46% Ni, and 20.72% MgO. In this ore, chalcopyrite is distributed in medium and fine uneven grains, and approximately 1–20% of pentlandite is distributed in pyrrhotite in fine grains. In order to efficiently develop and utilize such resources, a

bulk flotation process for copper and nickel was adopted to recover copper and nickel elements in the ore. When the grinding fineness is -0.074 mm (85%), through one roughing, two concentrating, two scavenging, and one concentrate scavenging operation, a qualified bulk copper-nickel concentrate with a copper grade of 3.96%, a copper recovery rate of 80.04%, a nickel grade of 5.71%, a nickel recovery rate of 81.50%, and 7.48% magnesium oxide was obtained.

Xiao et al. [71] conducted a research on the beneficiation process for improving the grade of copper concentrate and reducing magnesium in a copper-nickel ore dressing plant. In the experiment, an efficient talc inhibitor was used to suppress talc. Through one roughing and one scavenging operation, a copper concentrate with a copper grade of 24.68%, an MgO content of 2.12%, and a copper recovery rate of 69.55% was obtained. Compared with the original process, the grade of the copper concentrate increased by 2.47 percentage points, the copper recovery rate increased by 1.30 percentage points, and the content of the impurity MgO decreased by 2.96 percentage points. The new process has high production stability, stable technical indicators, and significant economic benefits, which have great reference significance for the separation of other similar ores.

Conclusions

Nickel is mainly derived from copper-nickel sulfide ore, silicate nickel ore, and laterite nickel ore. Laterite nickel ore and nickel silicate ore are both oxidized ores, which are not easy to separate by flotation, and are commonly treated by pyrometallurgy, hydrometallurgy, and leaching. They have a large investment, high production cost, high equipment consumption, and imperfect technology, and the application scale is small. Current global nickel resource development and utilization are given priority to copper-nickel sulfide ore. Copper-nickel sulfide ore composition is complex, the grade is low, it is easy to be oxidized, gangue ore argillization arises easily, and it has good floatability, low nickel recovery, higher magnesium gangue, and copper-nickel separation difficulties. So summarizing copper-nickel sulfide ore flotation separation technology is beneficial to reduce nickel metal loss, reduce the magnesium oxide content in concentrate, and improve the efficiency of copper and nickel separation.

With the continuous exploitation of copper-nickel sulfide ore, mineral resources are increasingly scarce. In order to effectively solve this problem, it is essential to continuously enhance the comprehensive utilization level of copper-nickel sulfide ore. Therefore, based on the process mineralogy characteristics of minerals, it is crucial to continuously develop high-performance flotation agents, optimize the process flow, and develop energy-saving and efficient mineral processing equipment.

Abbreviations

USGS	United States Geological Survey
CMC	Carboxymethyl cellulose
SHMP	Sodium hexametaphosphate
MIBC	Methyl isobutyl carbinol
PASP	Polyaspartic acid
PESA	Polyepoxysuccinic acid
ICP	Inductively coupled plasma

Acknowledgements

Not applicable.

Authors' contributions

XG was responsible for conceptualization, methodology, investigation, writing the original draft, data curation, formal analysis, validation, and review and editing. All authors read and approved the final manuscript.

Funding

No funding was received to assist with the preparation of this manuscript.

Data availability

Not applicable.

Declarations**Competing interests**

The authors declare that they have no competing interests.

Received: 9 October 2024 Accepted: 6 February 2025

Published online: 22 February 2025

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